Fe-C Problem 7 | Problems T4_PhaseDiagrams II.pdf #7 | Fe-C · hypereutectic cast iron

Hypereutectic Cast Iron from its Cementite Fraction

Original statement · Problems T4_PhaseDiagrams II.pdf, p. 2

7.- Consider an alloy of Fe-C, with content of C > 4.3%C at (727−ΔT)ºC. In these conditions, χ_{Fe_3C} = 0.73. Indicate the mass fraction of the phases at (1147+ΔT)ºC and (1147−ΔT)ºC.

Official Solution: T1: w_L = 0.75, w_{Fe_3C} = 0.25; T2: w_γ = 0.39, w_{Fe_3C} = 0.61, $w_{(Fe_3C)p}$ = 0.25, w_E = 0.75, $w_{\gamma E}$ = 0.39, $w_{(Fe_3C)E}$ = 0.36

PHASE 1 · HYPOTHESES & PARAMETERS

1. Equilibrium, Fe-Fe3C diagram [Session 9 Slide 4].
2. $C_0 > 4.3$: a **hypereutectic cast iron**; the primary phase is cementite, which forms in L + Fe3C above 1147 °C.
3. Unknown first: C_0 from the cementite fraction just below 727 °C.

PHASE 2 · FRAMES & STRATEGY

Frame: the Fe-Fe3C diagram (wt% C, °C). Tie lines: 727 − ΔT (α 0.022 / Fe3C 6.67), 1147 + ΔT (L 4.30 / Fe3C 6.67), 1147 − ΔT (γ 2.14 / Fe3C 6.67). Lever rule each time [Session 7 Slides 16, 32].

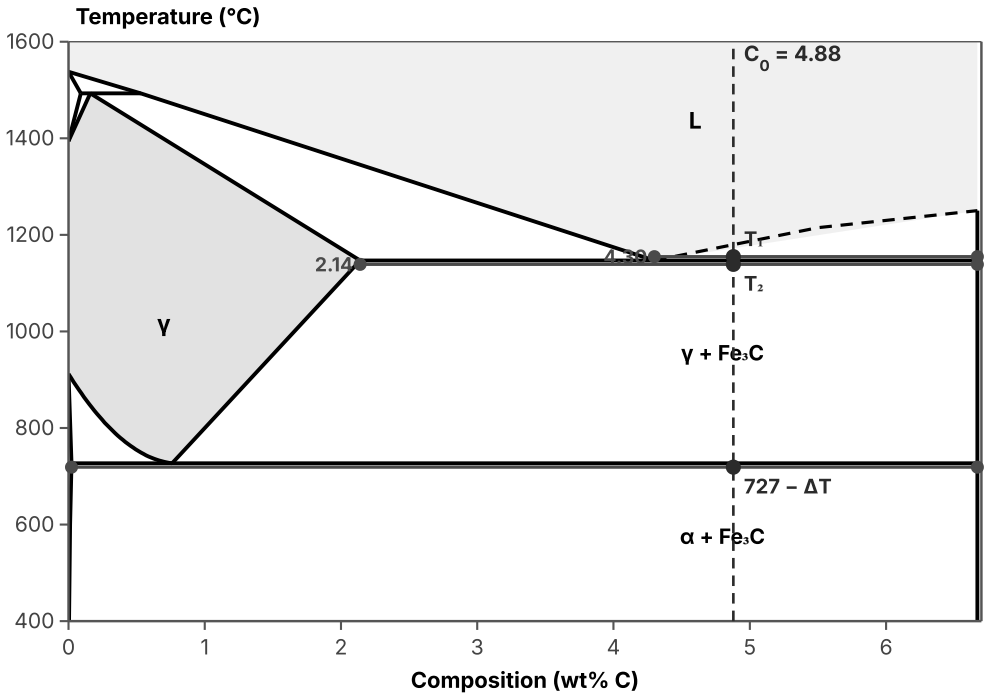


Figure 35. The alloy with 73 % Fe_3C just below 727 °C lies at about 4.88 % C: L + Fe_3C above 1147 °C, γ + Fe_3C below it.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Composition

$$C_0 = C_\alpha + W_{\mathrm{Fe_3C}}\,(C_{\mathrm{Fe_3C}} - C_\alpha) = 0.022 + 0.73 \times 6.648 = 0.022 + 4.853 = 4.875\ \%\,\mathrm{C}$$

2. T_1 **= 1147 + ΔT (length** $6.67 - 4.30 = 2.37$**)**

$$W_L = \frac{6.67 - 4.875}{2.37} = \frac{1.795}{2.37} = 0.757 \qquad W_{\mathrm{Fe_3C}} = \frac{4.875 - 4.30}{2.37} = \frac{0.575}{2.37} = 0.243$$

3. T_2 **= 1147 – ΔT (length** $6.67 - 2.14 = 4.53$**)**

$$W_\gamma = \frac{6.67 - 4.875}{4.53} = \frac{1.795}{4.53} = 0.396 \qquad W_{\mathrm{Fe_3C}} = \frac{4.875 - 2.14}{4.53} = \frac{2.735}{4.53} = 0.604$$

Microconstituents at T_2: primary cementite = 0.243; ledeburite = 0.757, made of γ 0.396 and Fe3C $0.604 - 0.243 = 0.361$.

PHASE 4 · CHECKS & INTERPRETATION

- Ledeburite make-up: $0.396/0.757 = 0.523$ γ (Worked example 14).
- Mass balance at T_1: $0.757 \times 4.30 + 0.243 \times 6.67 = 3.26 + 1.62 = 4.875$ % C .
- **Limiting cases** (fractions dimensionless, compositions in wt% C): at $C_0 = 4.30$ the primary Fe3C would be 0 and at $C_0 = 6.67$ the alloy would be all Fe3C; 4.875 % C lies near the eutectic end, hence 0.243 of primary Fe3C.

Check against the official solution

Agrees with the key in the eutectic Fe3C, 0.361 (key 0.36). The key does not give C_0; it is 4.875 % C (4.90 % with 6.7 % C for cementite: $0.022 + 0.73 \times 6.678 = 4.897$).

DISCREPANCY WITH THE OFFICIAL SOLUTION

Four values differ from the key by 0.01: 0.757/0.243 (key 0.75/0.25) and 0.396/0.604 (key 0.39/0.61). The key works with C_0 rounded to 4.9 % and 6.7 % C for cementite: $(6.7 - 4.9)/2.4 = 0.75$, $(4.9 - 4.3)/2.4 = 0.25$, $(6.7 - 4.9)/4.56 = 0.395 \to 0.39$ and $0.605 \to 0.61$. With the slides' 6.67 % C and the unrounded C_0 the values above follow; they are kept.